CSC3066
Deep Learning

Gradient Descent and Backpropagation for MLP

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Outline

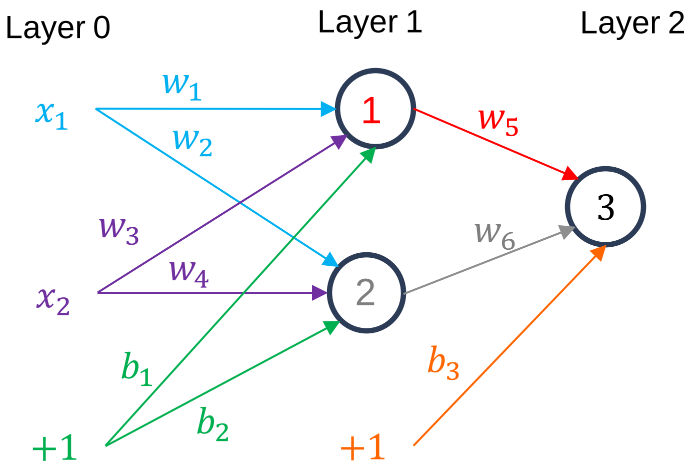
- Recap on MLP
- Training MLP with Gradient Descent and Backpropagation
- Implementing Gradient Descent for MLP:
  - Matrix notation for a single instance
  - Matrix notation for a batch of data

MLP:

Forward Pass

Backpropagation: MLP

- Let's consider a neural network with two inputs, two hidden neurons and one output neurons.
- We will use sigmoid activation function for each neuron

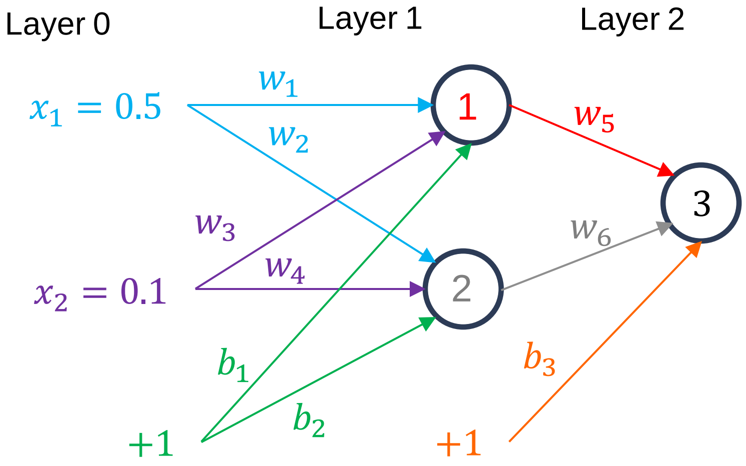


in_k - weighted sum of inputs of neuron k
out_k - output of neuron k
t – target output
σ – activation function

$$E = \frac{1}{2}\,(t - out_3)^2$$

Backpropagation: MLP

- Let's consider the training example from the last lecture: given inputs $\mathbf{0.5}$ and $\mathbf{0.1}$, we want the network to output $\mathbf{0.9}$.

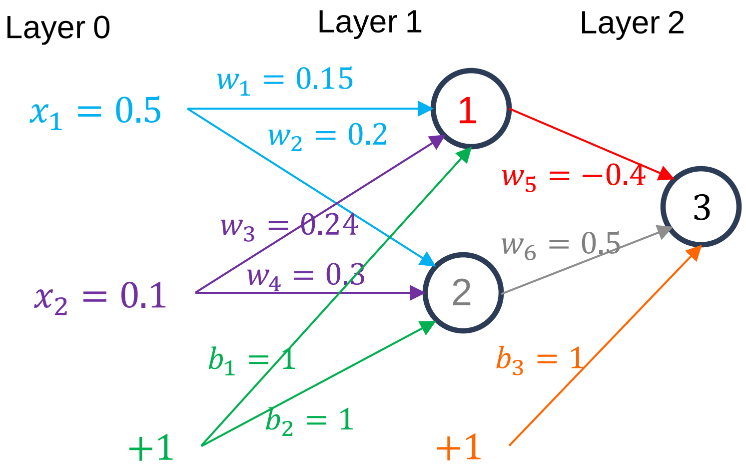


in_k - weighted sum of inputs of neuron k
out_k - output of neuron k
t – target output
σ – activation function

$$E = \frac{1}{2}(t - out_3)^2$$

Backpropagation: MLP

- We initialise weights and biases with some random values.



in_k - weighted sum of inputs of neuron k
out_k - output of neuron k
t – target output
σ – activation function

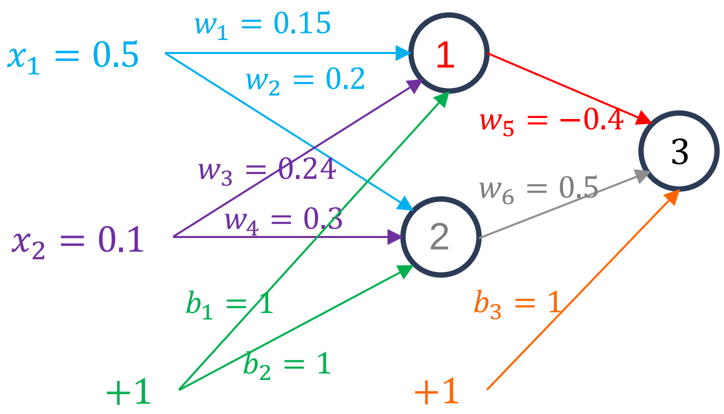
$$E = \frac{1}{2}(t - out_3)^2$$

Backpropagation: MLP

- Forward Pass (Layer 1):

$$in_1 = x_1 w_1 + x_2 w_3 + b_1 = 1.099 \qquad out_1 = \sigma(in_1) = \frac{1}{1+e^{-(in_1)}} = 0.75$$

$$in_2 = x_1 w_2 + x_2 w_4 + b_2 = 1.13 \qquad out_2 = \sigma(in_2) = \frac{1}{1+e^{-(in_2)}} = 0.756$$

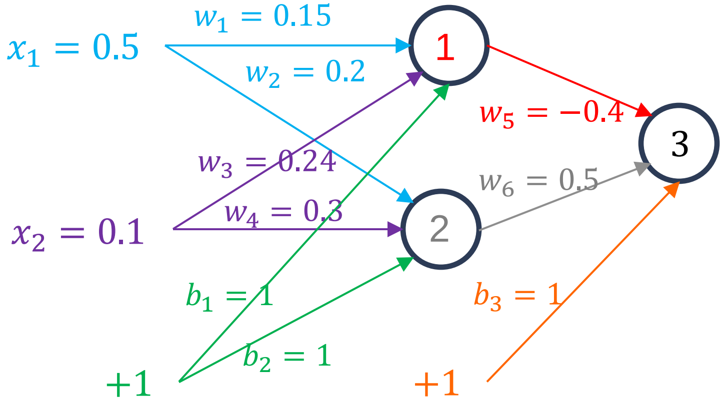


Backpropagation: MLP

- Forward Pass (Layer 2):

$$in_3 = out_1 w_5 + out_2 w_6 + b_3 = 1.078$$

$$out_3 = \frac{1}{1 + e^{-(in_3)}} = 0.75$$



Backpropagation: MLP

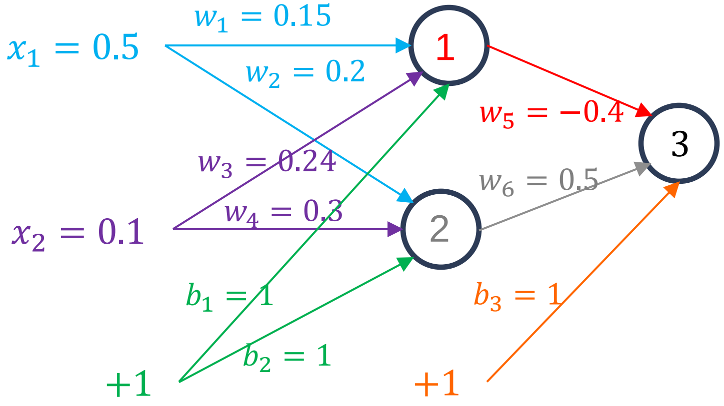
- Forward Pass (matrix notation):

Exercise:

$X =$

$W^1 =$ $B^1 =$

$W^2 =$ $B^2 =$



Backpropagation: MLP

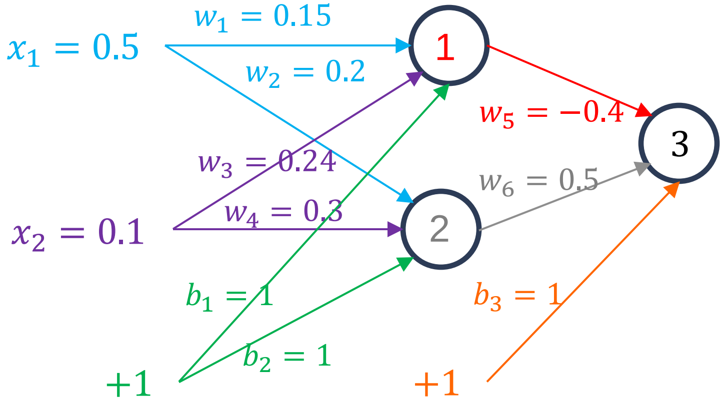
- Forward Pass (matrix notation):

$X = [x_1, x_2] = [0.5, 0.1]$

$W^1 = \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \end{bmatrix} = \begin{bmatrix} 0.15 & 0.2 \\ 0.24 & 0.3 \end{bmatrix}$ $B^1 = [b_1, b_2] = [1, 1]$

$W^2 = \begin{bmatrix} w_5 \\ w_6 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.5 \end{bmatrix}$ $B^2 = [b_3] = [1]$

Exercise: Show that
$\sigma\left(\sigma(XW^1 + B^1)W^2 + B^2\right) = 0.75$



Backpropagation: MLP

- Forward Pass (matrix notation):

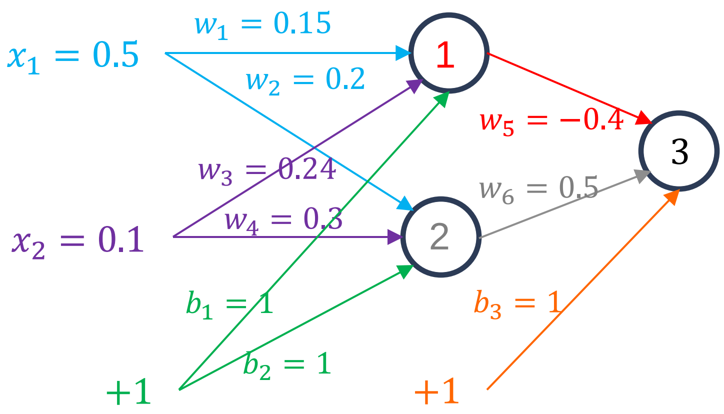
$X = [x_1, x_2] = [0.5, 0.1]$

$W^1 = \begin{bmatrix} w_1 & w_2 \\ w_3 & w_4 \end{bmatrix} = \begin{bmatrix} 0.15 & 0.2 \\ 0.24 & 0.3 \end{bmatrix}$ $B^1 = [b_1, b_2] = [1,1]$

$W^2 = \begin{bmatrix} w_5 \\ w_6 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.5 \end{bmatrix}$ $B^2 = [b_3] = [1]$

Exercise: Show that
$\sigma\,(\sigma(XW^1 + B^1)W^2 + B^2) = 0.75$

In^k- input to layer k (matrix notation)
Out^k- output of layer k (matrix notation)



$In^1 = XW^1 + B^1 = [1.099, 1.13]$ - input of Layer 1

$Out^1 = \sigma(In^1) = \sigma[in_1, in_2] = [0.75, 0.756]$ - output of Layer 1

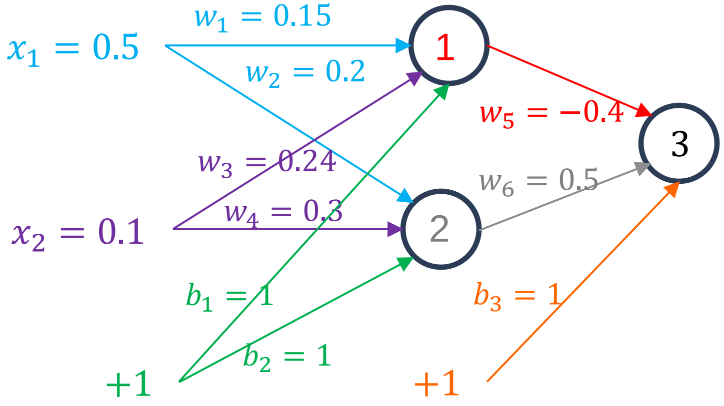
$In^2 = Out^1W^2 + B^2 = 1.078$ - input of Layer 2

$Out^2 = \sigma(In^2) = 0.75$ - output of Layer 2

Backpropagation: MLP

- Error Calculation:

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}(0.9 - 0.75)^2 = 0.0113$$



MLP:

Backpropagation through Layer 2

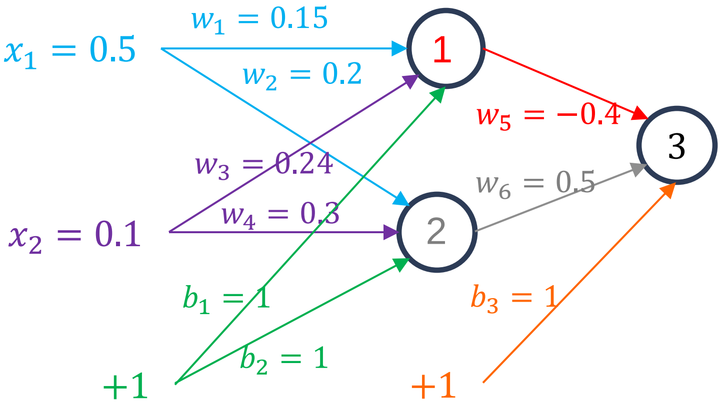
Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_3}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$

$$\frac{\partial E}{\partial w_5} =$$

Exercise: Use the chain rule to write formula for $\frac{\partial E}{\partial w_5}$

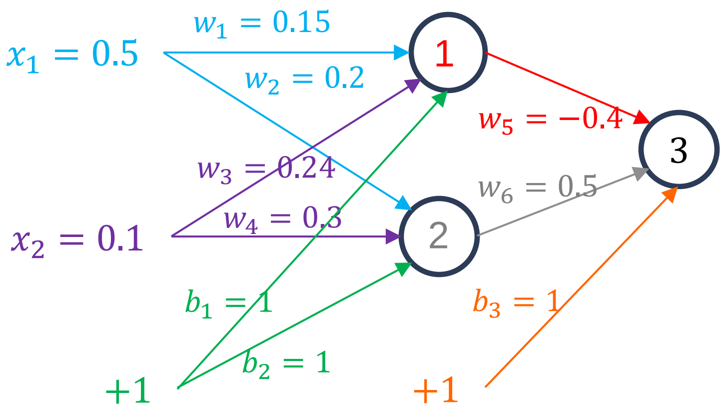


Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_3}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$

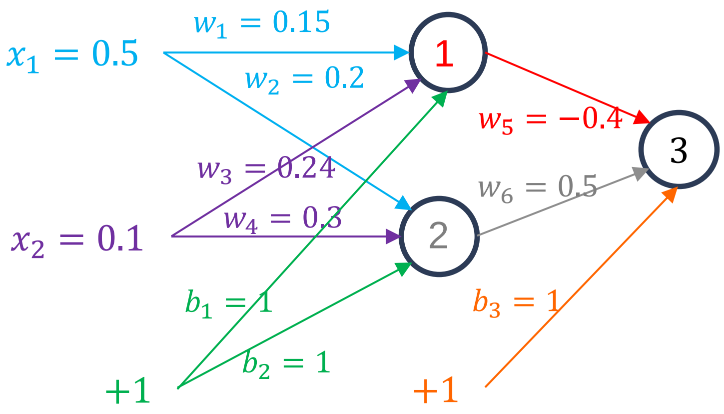
$$\frac{\partial E}{\partial w_5} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial w_5}$$



Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_3}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$



$$\frac{\partial E}{\partial w_5} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial w_5}$$

$$\frac{\partial E}{\partial out_3} =$$

$$\frac{\partial out_3}{\partial in_3} =$$

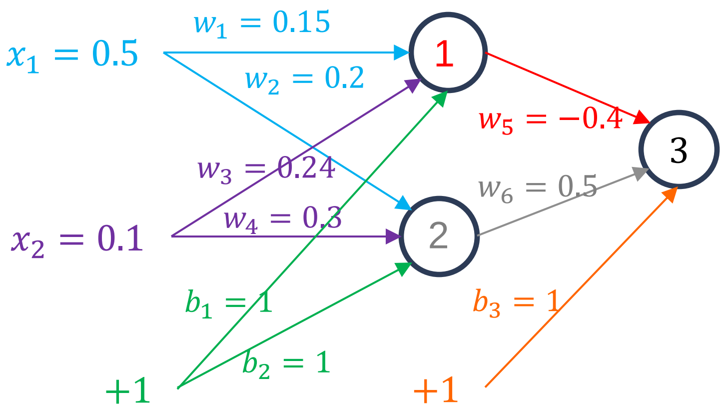
$$\frac{\partial in_3}{\partial w_5} =$$

Exercise: Calculate the derivatives

Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_3}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$



$$\frac{\partial E}{\partial w_5} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial w_5} =$$

$$\frac{\partial E}{\partial out_3} = out_3 - t =$$

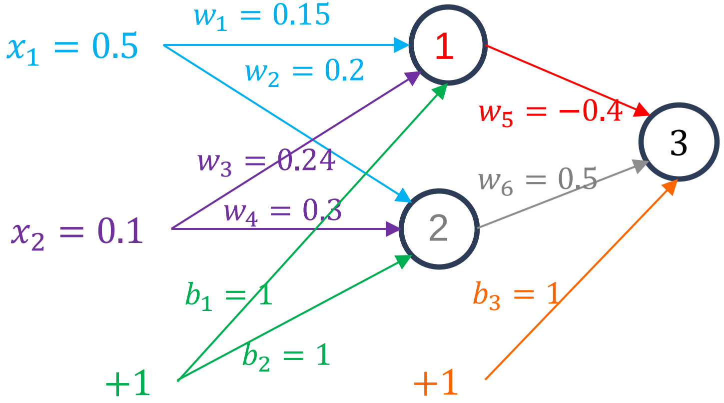
$$\frac{\partial out_3}{\partial in_3} = \frac{1}{1+e^{-in_3}}\left(1 - \frac{1}{1+e^{-in_3}}\right) =$$

$$\frac{\partial in_3}{\partial w_5} = out_1 =$$

Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_3}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$



$$\frac{\partial E}{\partial w_5} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial w_5} = -0.022$$

$$\frac{\partial E}{\partial out_3} = out_3 - t = -0.154$$

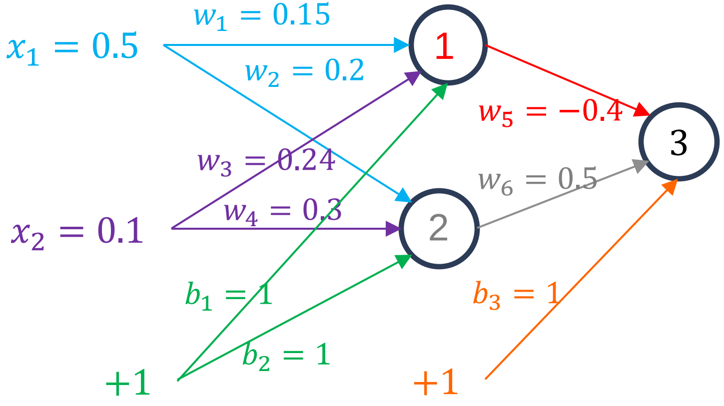
$$\frac{\partial out_3}{\partial in_3} = \frac{1}{1+e^{-in_3}}\left(1 - \frac{1}{1+e^{-in_3}}\right) = 0.189$$

$$\frac{\partial in_3}{\partial w_5} = out_1 = 0.75$$

Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_1}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$



$$\frac{\partial E}{\partial w_6} = \frac{\partial E}{\partial out_3} \frac{\partial out_3}{\partial in_3} \frac{\partial in_3}{\partial w_6} = -0.022$$

$$\frac{\partial E}{\partial out_3} = out_3 - t = -0.15$$

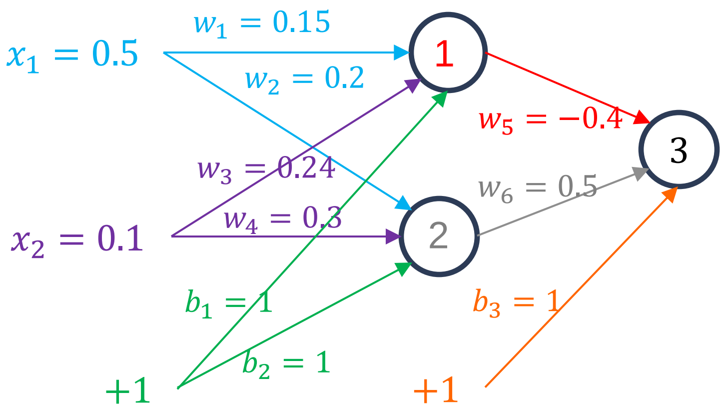
$$\frac{\partial out_3}{\partial in_3} = \frac{1}{1+e^{-in_3}}\left(1 - \frac{1}{1+e^{-in_3}}\right) = 0.1895$$

$$\frac{\partial in_3}{\partial w_6} = out_2 = 0.756$$

Backpropagation: MLP

- Now we need to calculate $\frac{\partial E}{\partial w_i}$ for $w_1, \cdots, w_6$ and $\frac{\partial E}{\partial b_i}$ for b_1, b_2, b_3
- From previous example we already know how to calculate $\frac{\partial E}{\partial w_5}, \frac{\partial E}{\partial w_6}, \frac{\partial E}{\partial b_1}$

$$E = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}\left(t - \sigma(in_3)\right)^2 = \frac{1}{2}\left(t - \sigma(w_5 \times out_1 + w_6 \times out_2 + b_3)\right)^2$$



$$\frac{\partial E}{\partial b_3} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial b_3} = -0.029$$

$$\frac{\partial E}{\partial out_3} = out_3 - t = -0.15$$

$$\frac{\partial out_3}{\partial in_3} = \frac{1}{1+e^{-in_3}}\left(1 - \frac{1}{1+e^{-in_3}}\right) = 0.1895$$

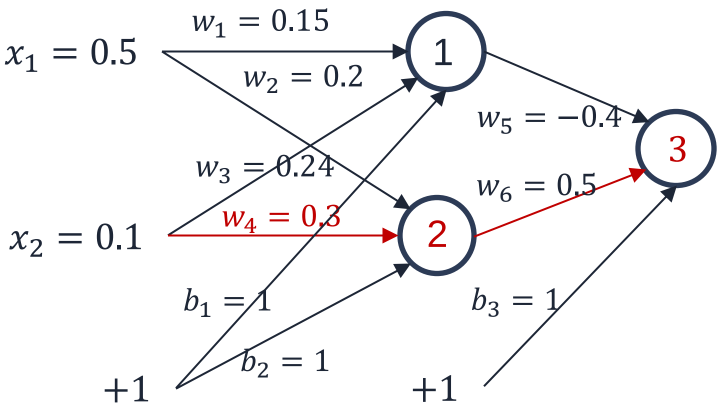
$$\frac{\partial in_3}{\partial b_3} = 1$$

MLP:

Backpropagation through Layer 1

Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$



$$E = \frac{1}{2}(y - out_3)^2$$

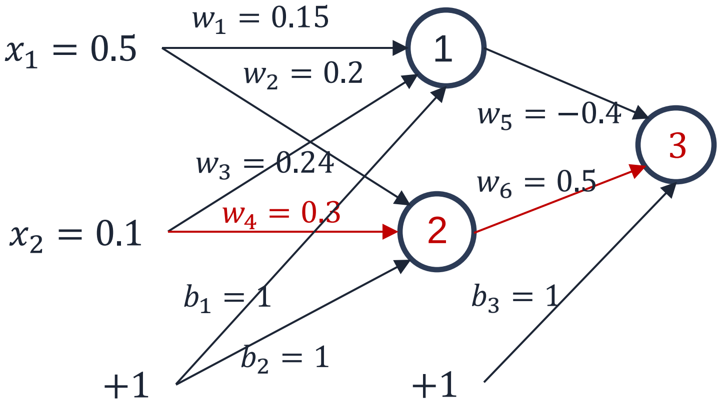
$$out_3 = \sigma(in_3) = \sigma(w_5 out_1 + w_6 out_2 + b_3) =$$

$$\sigma(w_5\sigma(in_1) + w_6\sigma(in_2) + b_3) =$$

$$\sigma(w_5\sigma(x_1w_1 + x_2w_3 + b_1) + w_6\sigma(x_1w_2 + x_2w_4 + b_2) + b_3)$$

Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$



$$E = \frac{1}{2}(y - out_3)^2$$

$$out_3 = \sigma(in_3) = \sigma(w_5 out_1 + w_6 out_2 + b_3) =$$

$$\sigma(w_5 \sigma(in_1) + w_6 \sigma(in_2) + b_3) =$$

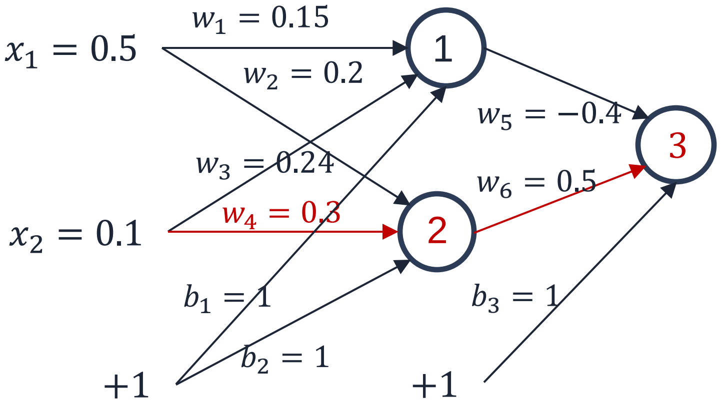
$$\sigma(w_5 \sigma(x_1 w_1 + x_2 w_3 + b_1) + w_6 \sigma(x_1 w_2 + x_2 w_4 + b_2) + b_3)$$

$$\frac{\partial E}{\partial w_4} = ?$$

Exercise: Use the chain rule to write the formula for $\frac{\partial E}{\partial w_4}$

Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$



$$E = \frac{1}{2}(y - out_3)^2$$

$$out_3 = \sigma(in_3) = \sigma(w_5 out_1 + w_6 out_2 + b_3) =$$

$$\sigma(w_5\sigma(in_1) + w_6\sigma(in_2) + b_3) =$$

$$\sigma(w_5\sigma(x_1w_1 + x_2w_3 + b_1) + w_6\sigma(x_1w_2 + x_2w_4 + b_2) + b_3)$$

$$\frac{\partial E}{\partial w_4} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_2}\frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial w_4}$$

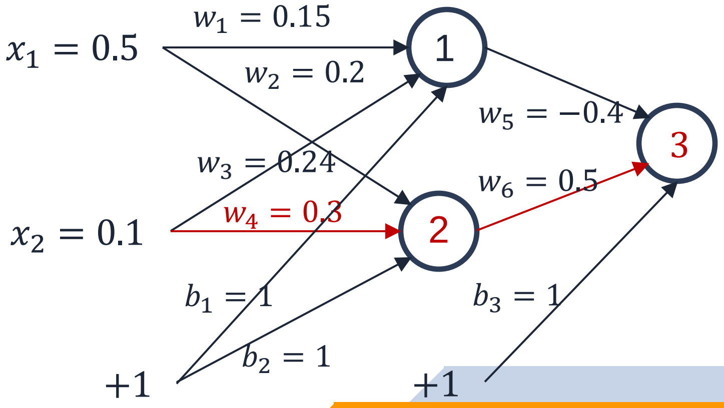
Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$

$$\frac{\partial E}{\partial w_4} = \underbrace{\boxed{\frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}}}_{-0.15 \times 0.1895}\underbrace{\boxed{\frac{\partial in_3}{\partial out_2}}}_{0.5}\frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial w_4}$$

We calculated this earlier

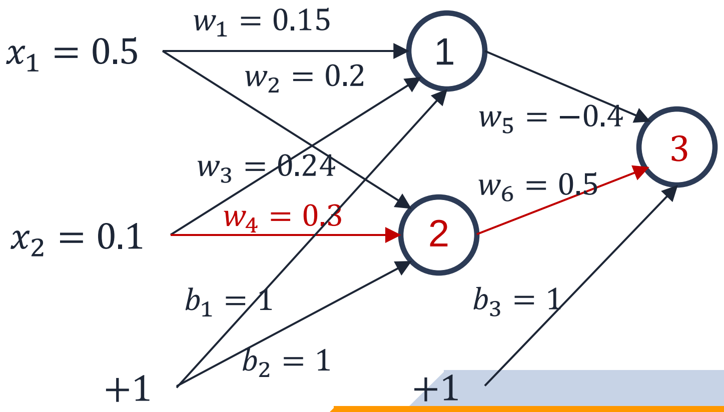
This is equal to w_6



Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$

$$\frac{\partial E}{\partial w_4} = -0.15 \times 0.1895 \times 0.5 \times \frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial w_4}$$



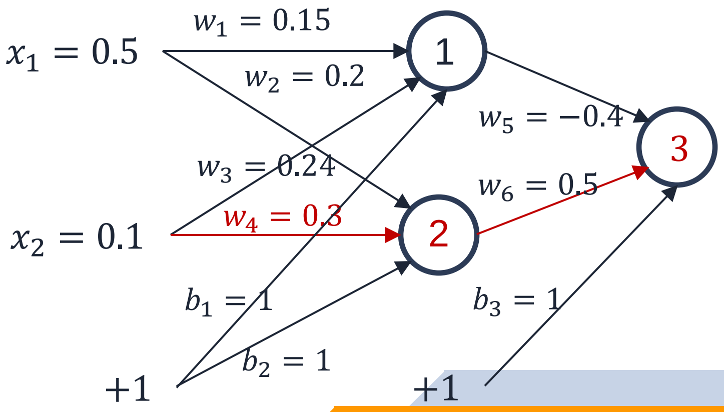
Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$?

$$\frac{\partial E}{\partial w_4} = -0.0142 \times \frac{\partial out_2}{\partial in_2} \times \frac{\partial in_2}{\partial w_4}$$

$$\frac{\partial out_2}{\partial in_2} = ?$$

$$\frac{\partial in_2}{\partial w_4} = ?$$



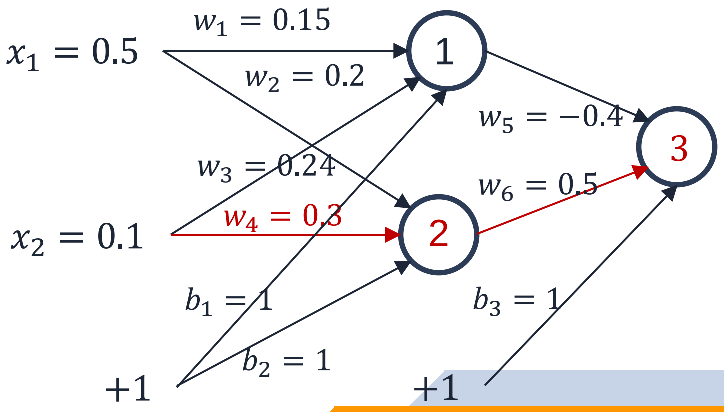
Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$?

$$\frac{\partial E}{\partial w_4} = -0.0142 \times \frac{\partial out_2}{\partial in_2} \times \frac{\partial in_2}{\partial w_4}$$

$$\frac{\partial out_2}{\partial in_2} = 0.756 \times (1 - 0.756) = 0.1845$$

$$\frac{\partial in_2}{\partial w_4} =?$$



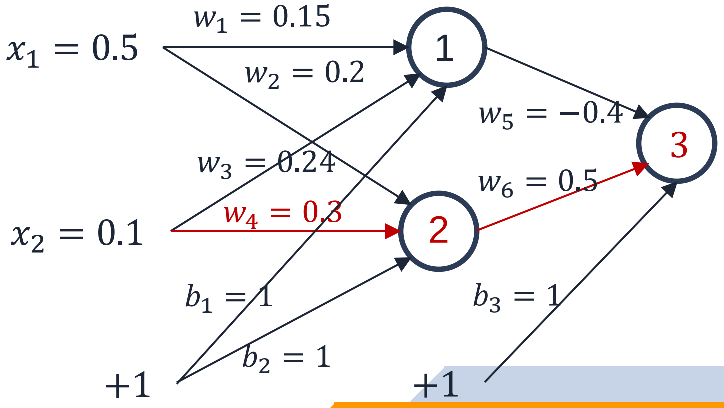
Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$?

$$\frac{\partial E}{\partial w_4} = -0.0142 \times \frac{\partial out_2}{\partial in_2} \times \frac{\partial in_2}{\partial w_4}$$

$$\frac{\partial out_2}{\partial in_2} = 0.756 \times (1 - 0.756) = 0.1845$$

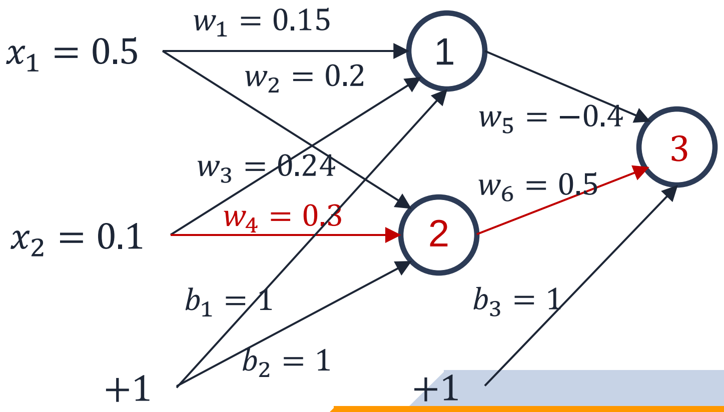
$$\frac{\partial in_2}{\partial w_4} = x_2 = 0.1$$



Backpropagation: MLP

- What about $\frac{\partial E}{\partial w_4}$?

$$\frac{\partial E}{\partial w_4} = -0.0142 \times 0.1845 \times 0.1$$



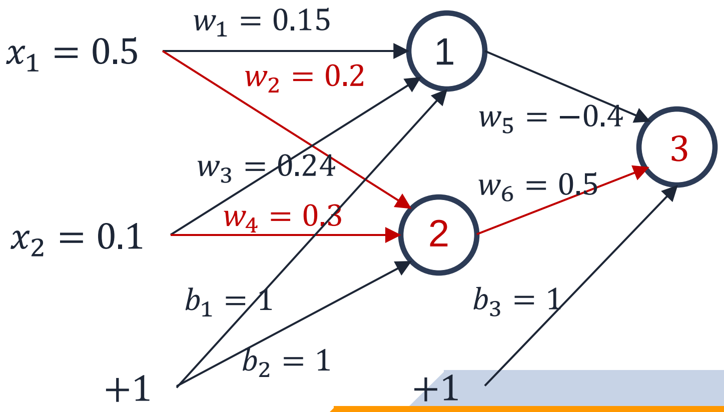
Backpropagation: MLP

$$\frac{\partial E}{\partial w_4} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_2}\frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial w_4}$$

$$\frac{\partial E}{\partial w_4} = -0.0142 \times 0.1845 \times 0.1 = -0.0003$$

$$\frac{\partial E}{\partial w_2} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_2}\frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial w_2}$$

$$\frac{\partial E}{\partial w_2} = -0.0142 \times 0.1845 \times 0.5 = -0.0013$$

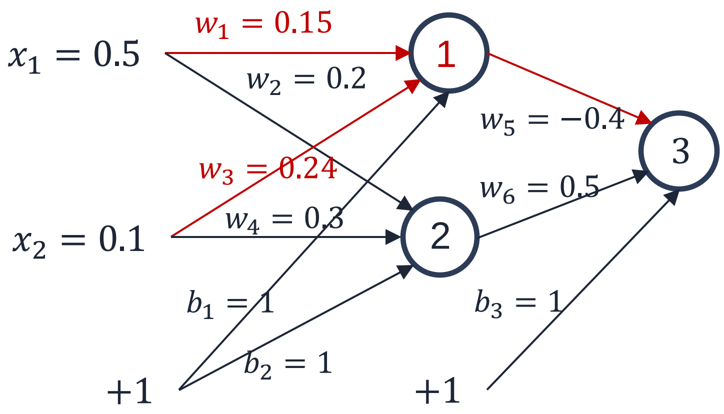


Backpropagation: MLP

$$\frac{\partial E}{\partial w_1} =$$

Exercise: Use the chain rule to write the formulas for $\frac{\partial E}{\partial w_1}$ and $\frac{\partial E}{\partial w_3}$

$$\frac{\partial E}{\partial w_3} =$$



$$E = \frac{1}{2}(y - out_3)^2$$

$$out_3 = \sigma(in_3) = \sigma(w_5 out_1 + w_6 out_2 + b_3) =$$

$$\sigma(w_5\sigma(in_1) + w_6\sigma(in_2) + b_3) =$$

$$\sigma(w_5\sigma(x_1 w_1 + x_2 w_3 + b_1) + w_6\sigma(x_1 w_2 + x_2 w_4 + b_2) + b_3)$$

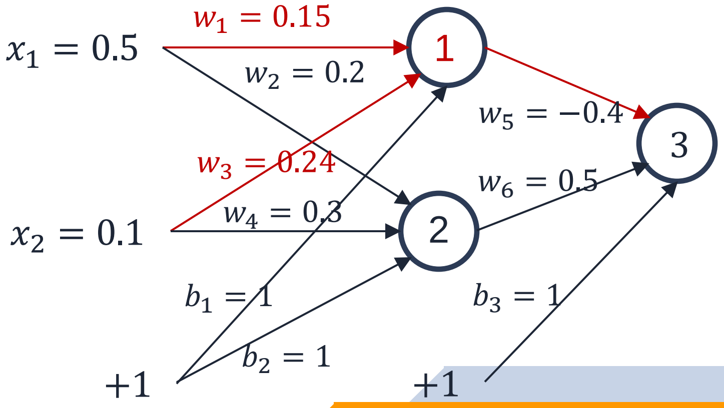
Backpropagation: MLP

$$\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_1}\frac{\partial out_1}{\partial in_1}\frac{\partial in_1}{\partial w_1}$$

$$\frac{\partial E}{\partial w_1} = 0.0011$$

$$\frac{\partial E}{\partial w_3} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_1}\frac{\partial out_1}{\partial in_1}\frac{\partial in_1}{\partial w_3}$$

$$\frac{\partial E}{\partial w_3} = 0.0002$$



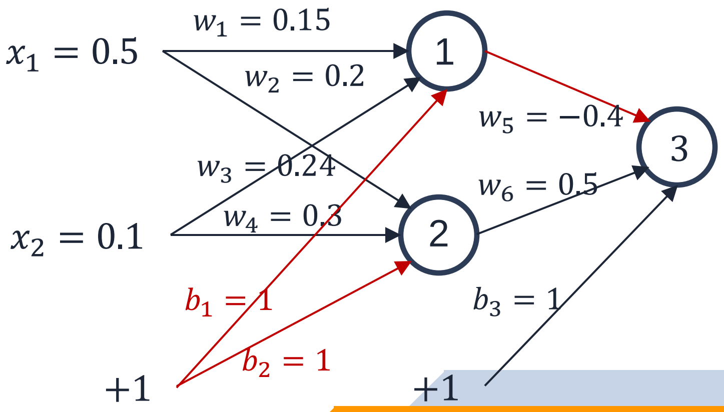
Backpropagation: MLP

$$\frac{\partial E}{\partial b_1} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_1}\frac{\partial out_1}{\partial in_1}\frac{\partial in_1}{\partial b_1}$$

$$\frac{\partial E}{\partial b_1} = 0.0022$$

$$\frac{\partial E}{\partial b_2} = \frac{\partial E}{\partial out_3}\frac{\partial out_3}{\partial in_3}\frac{\partial in_3}{\partial out_2}\frac{\partial out_2}{\partial in_2}\frac{\partial in_2}{\partial b_2}$$

$$\frac{\partial E}{\partial b_2} = -0.0027$$



Backpropagation: MLP

$$\frac{\partial E}{\partial w_1} = 0.0011 \qquad \frac{\partial E}{\partial w_2} = -0.0013$$

$$\frac{\partial E}{\partial w_3} = -0.0002 \qquad \frac{\partial E}{\partial w_4} = -0.0003$$

$$\frac{\partial E}{\partial w_5} = -0.022 \qquad \frac{\partial E}{\partial w_6} = -0.022$$

$$\frac{\partial E}{\partial b_1} = 0.0022 \qquad \frac{\partial E}{\partial b_2} = -0.0027 \qquad \frac{\partial E}{\partial b_3} = -0.029$$

Backpropagation: MLP

- Now we can update the weights: $w_i^{(new)} = w_i - \alpha \frac{\partial E}{\partial w_i}$

$$w_1^{(new)} = 0.15 - 0.5 \times (0.0011) = 0.149$$

$$w_2^{(new)} = 0.2 - 0.5 \times (-0.0013) = 0.239$$

$$w_3^{(new)} = 0.24 - 0.5 \times (-0.0002) = 0.2$$

$$w_4^{(new)} = 0.3 - 0.5 \times (-0.0003) = 0.3001$$

$$w_5^{(new)} = -0.4 - 0.5 \times (-0.022) = -0.389$$

$$w_6^{(new)} = 0.5 - 0.5 \times (-0.022) = 0.511$$

Backpropagation: MLP

- We do the same with bias: $b_i^{(new)} = b_i - \alpha \frac{\partial E}{\partial b_i}$

$$b_1^{(new)} = 1 - 0.5 \times (0.0022) = 0.999$$

$$b_2^{(new)} = 1 - 0.5 \times (-0.0027) = 1.0013$$

$$b_3^{(new)} = 1 - 0.5 \times (-0.029) = 1.0146$$

Backpropagation: MLP

- Error Calculation:

$$in_1 = x_1w_1 + x_2w_3 + b_1 = 1.0985$$

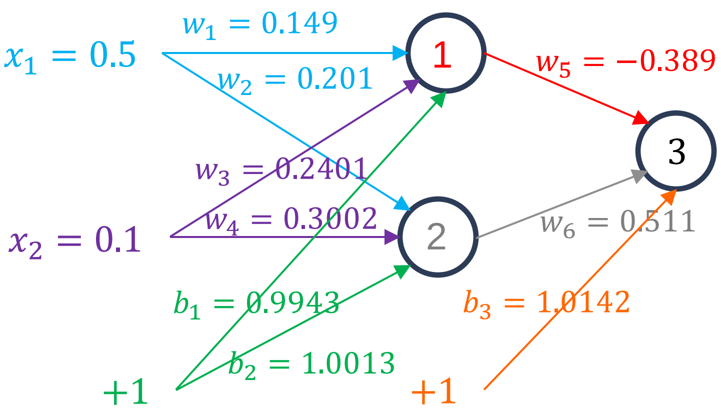
$$in_2 = x_1w_2 + x_2w_4 + b_2 = 1.1305$$

$$in_3 = out_1w_5 + out_2w_6 + b_3 = 1.1087$$

$$out_1 = \frac{1}{1+e^{-(n_1)}} = 0.74998$$

$$out_2 = \frac{1}{1+e^{-(in_2)}} = 0.75593$$

$$out_3 = \frac{1}{1+e^{-(in_3)}} = 0.75189$$



$$E^0 = 0.013$$

$$E^1 = \frac{1}{2}(t - out_3)^2 = \frac{1}{2}(0.9 - 0.752)^2 = 0.011$$

Backproparadion: MLP matrix notation

- How can we implement it?
- The idea behind Backpropagation is to calculate all the gradients at the same time rather than one by one
- We need a matrix notation…

MLP:

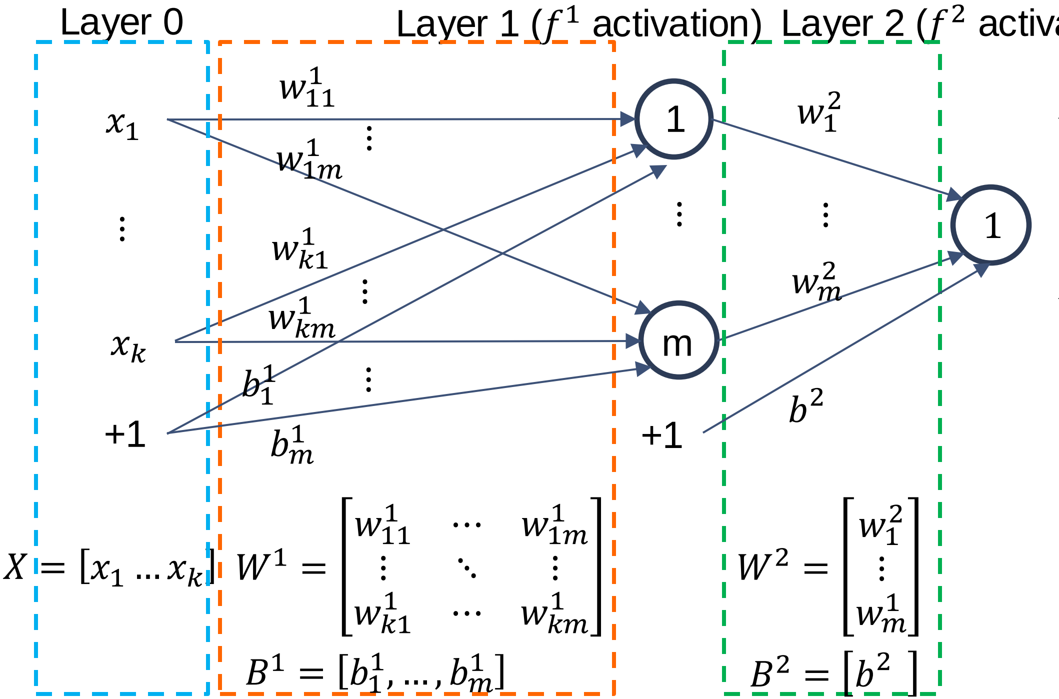
Implementing Gradient Descent using Matrix Notation

MLP:

Matrix Notation: Single Instance

Backproparadion: MLP matrix notation

- Matrix notation:



$$In^1 = XW^1 + B^1 = [in_1^1 \dots in_m^1]$$

$$Out^1 = [out_1^1, \cdots, out_m^1]$$

$$In^2 = Out^1 W^2 + B^2 = in_1^2$$

$$Out^2 = [out_1^2]$$

MLP:

Matrix Notation: Single Instance

LAYER 2

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \dots, x_k]$)

$$\frac{\partial E_x}{\partial W^2} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial w_1^2} \\ \vdots \\ \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial w_m^2} \end{bmatrix}$$

$\frac{\partial E}{\partial w_1^2}$ $\quad out_1^1$

$\frac{\partial E}{\partial w_m^2}$ $\quad out_m^1$

$$\frac{\partial E_x}{\partial B^2} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial b^2} \end{bmatrix}$$

$\frac{\partial E}{\partial b^2}$ $\quad 1$

$$W^2 = \begin{bmatrix} w_1^2 \\ \vdots \\ w_m^2 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} b^2 \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \ldots, x_k]$)

$$\frac{\partial E_x}{\partial W^2} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} out_1^1 \\ \vdots \\ \frac{\partial E_1}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} out_m^1 \end{bmatrix} = {Out^1}^T \left(\frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} \right)$$

scalar

matrix[no of neurons in Layer 1, 1]

scalar

$$W^2 = \begin{bmatrix} w_1^2 \\ \vdots \\ w_m^2 \end{bmatrix}$$

$$Out^1 = [out_1^1, \cdots, out_m^1]$$

Backproparadion: definition

Matrix notation (single instance $x = [x_1, \ldots, x_k]$) :

scalar

$$\frac{\partial E_x}{\partial B^2} = 1 \times \left(\frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} \right)$$

scalar

$B^2 = [b^2]$

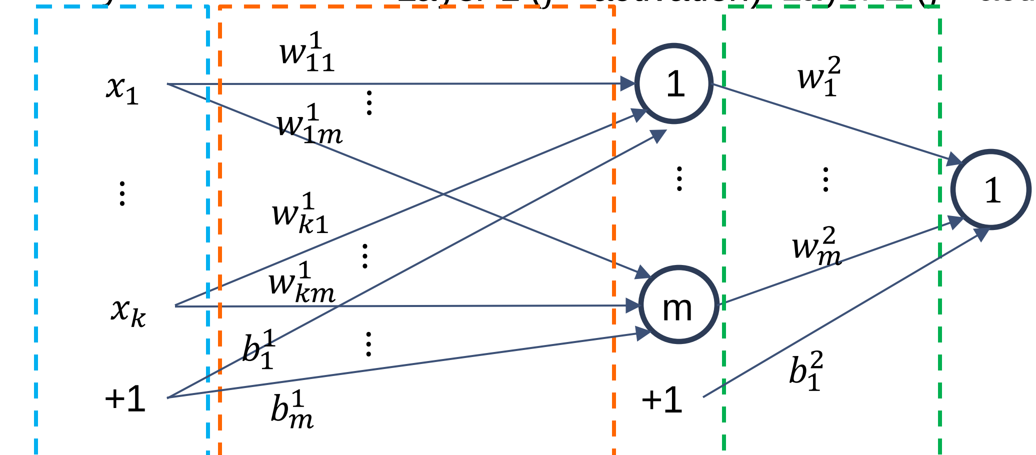
MLP:

Matrix Notation: Single Instance

LAYER 1

- Matrix notation:

Layer 0 — Layer 1 (f^1 activation) — Layer 2 (f^2 activation)



$$X = [x_1 \dots x_k] \quad W^1 = \begin{bmatrix} w_{11}^1 & \cdots & w_{1m}^1 \\ \vdots & \ddots & \vdots \\ w_{k1}^1 & \cdots & w_{km}^1 \end{bmatrix} \quad W^2 = \begin{bmatrix} w_1^2 \\ \vdots \\ w_m^1 \end{bmatrix}$$

$$B^1 = [b_1^1, \dots, b_m^1] \quad B^2 = [b_1^2]$$

$$In^1 = XW^1 + B^1 = [in_1^1 \dots in_m^1]$$

$$Out^1 = [out_1^1, \cdots, out_m^1]$$

$$In^2 = Out^1 W^2 + B^2 = in_1^2$$

$$Out^2 = [out_1^2]$$

Gradient Optimisation Method: Example

in_1^1 ← layer ← neuron

Matrix notation (single instance $x = [x_1, \ldots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial w_{11}^1} & \cdots & \frac{\partial E}{\partial w_{1m}^1} \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial w_{k1}^1} & \cdots & \frac{\partial E}{\partial w_{km}^1} \end{bmatrix}$$

$$W^1 = \underbrace{\begin{bmatrix} w_{11} & \cdots & w_{1m} \\ \cdots & \ddots & \cdots \\ w_{k1} & \cdots & w_{km} \end{bmatrix}}_{\text{Number of neurons}} \left.\vphantom{\begin{bmatrix} a \\ b \\ c \end{bmatrix}}\right\} \text{Number of inputs}$$

Gradient Optimisation Method: Example

in_1^1 ← layer
← neuron

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E}{\partial w_{11}^1} \qquad \frac{\partial E}{\partial w_{km}^1}$$

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1}\frac{\partial in_1^1}{w_{11}^1} & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1}\frac{\partial in_m^1}{w_{1m}^1} \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1}\frac{\partial in_1^1}{w_{k1}^1} & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1}\frac{\partial in_m^1}{w_{km}^1} \end{bmatrix}$$

$$W^1 = \underbrace{\begin{bmatrix} w_{11} & \cdots & w_{1m} \\ \cdots & \ddots & \cdots \\ w_{k1} & \cdots & w_{km} \end{bmatrix}}_{\text{Number of neurons}} \left.\vphantom{\begin{bmatrix} w \\ w \\ w \end{bmatrix}}\right\} \text{Number of inputs}$$

Gradient Optimisation Method: Example

in_1^1 layer neuron

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{array}{c} \left[\begin{array}{ccc} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1}\frac{\partial in_1^1}{w_{11}^1} & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1}\frac{\partial in_m^1}{w_{1m}^1} \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1}\frac{\partial in_1^1}{w_{k1}^1} & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1}\frac{\partial in_m^1}{w_{km}^1} \end{array}\right] \end{array} \begin{array}{c} x_1 \\ \\ x_k \end{array}$$

Gradient Optimisation Method: Example

in^{1}_{1} ← layer
← neuron

Matrix notation (single instance $x = [x_1, \ldots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1} x_1 & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1} x_1 \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1} x_k & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1} x_k \end{bmatrix}$$

Gradient Optimisation Method: Example

in_1^1 ← layer
← neuron

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1} x_1 & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1} x_1 \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1} x_k & \cdots & \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1} x_k \end{bmatrix}$$

w_1^2 w_m^2

Gradient Optimisation Method: Example

in_1^1 layer neuron

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_1^2 \frac{\partial out_1^1}{\partial in_1^1} x_1 & \cdots & \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_m^2 \frac{\partial out_m^1}{\partial in_m^1} x_1 \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_1^2 \frac{\partial out_1^1}{\partial in_1^1} x_k & \cdots & \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_m^2 \frac{\partial out_m^1}{\partial in_m^1} x_k \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E_x}{\partial W^1} = \begin{bmatrix} \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_1^2 \frac{\partial out_1^1}{\partial in_1^1} x_1 & \cdots & \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_m^2 \frac{\partial out_m^1}{\partial in_m^1} x_1 \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_1^2 \frac{\partial out_1^1}{\partial in_1^1} x_k & \cdots & \frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} w_m^2 \frac{\partial out_m^1}{\partial in_m^1} x_k \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \ldots, x_k]$):

The matrix from the previous slide can be obtained using the formula:

Exercise: Calculate manually to demonstrate (homework)

$$\frac{\partial E_x}{\partial W^1} = x^T \times \left(\left(\frac{\partial E}{\partial Out^2} * \frac{\partial Out^2}{\partial In^2} \times {W^2}^T \right) * \frac{\partial Out^1}{\partial In^1} \right)$$

matrix [1,no of neurons in Layer 1]

matrix [k,1]

matrix [no of neurons in Layer 2, no of neurons in Layer 1]

matrix [1, no of neurons in Layer 2]

matrix [1,no of neurons in Layer 2]

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

The matrix from the previous slide can be obtained using the formula:

$$\frac{\partial E_x}{\partial W^1} = x^T \times \left(\left(\frac{\partial E}{\partial Out^2} * \frac{\partial Out^2}{\partial In^2} \times {W^2}^T \right) * \frac{\partial Out^1}{\partial In^1} \right)$$

$$x^T = \begin{bmatrix} x_1 \\ \cdot \\ \cdot \\ \cdot \\ x_k \end{bmatrix} \qquad \frac{\partial Out^1}{\partial In^1} = \begin{bmatrix} \frac{\partial out_1^1}{\partial in_1^1} & \dots & \frac{\partial out_m^1}{\partial in_m^1} \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \dots, x_k]$):

$$\frac{\partial E_x}{\partial B^1} = \left[\overbrace{\frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} \frac{\partial in_1^2}{\partial out_1^1} \frac{\partial out_1^1}{\partial in_1^1} \frac{\partial in_1^1}{b_1^1}}^{\frac{\partial E}{\partial b_1^1}}, \dots, \overbrace{\frac{\partial E}{\partial out_1^2} \frac{\partial out_1^2}{\partial in_1^2} \frac{\partial in_1^2}{\partial out_m^1} \frac{\partial out_m^1}{\partial in_m^1} \frac{\partial in_m^1}{b_m^1}}^{\frac{\partial E}{\partial b_m^1}} \right]$$

Gradient Optimisation Method: Example

Matrix notation (single instance $x = [x_1, \ldots, x_k]$):

$$\frac{\partial E_x}{\partial B^1} = \left[\frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_1^1}\frac{\partial out_1^1}{\partial in_1^1}\frac{\partial in_1^1}{b_1^1}, \ldots, \frac{\partial E}{\partial out_1^2}\frac{\partial out_1^2}{\partial in_1^2}\frac{\partial in_1^2}{\partial out_m^1}\frac{\partial out_m^1}{\partial in_m^1}\frac{\partial in_m^1}{b_m^1}\right]$$

$$\frac{\partial E_x}{\partial B^1} = \left(\frac{\partial E}{\partial Out^2} * \frac{\partial Out^2}{\partial In^2} \times {W^2}^T\right) * \frac{\partial Out^1}{\partial In^1}$$

matrix [1,no of neurons in Layer 1]

matrix [no of neurons in Layer 2, no of neurons in Layer 1]

matrix [1,no of neurons in Layer 2]

Python Programming Example

python

MLP Backpropagation.ipynb

MLP:

Matrix Notation: Batch

MLP:

Matrix Notation: Batch

LAYER 2

Backproparadion: MLP matrix notation

Matrix notation (batch): $X = (x^1 \ldots x^n)$ where: $x^i = \left(x_1^i, \ldots, x_k^i\right)$

$$\frac{\partial E_X}{\partial W^2} = \begin{bmatrix} \frac{1}{n}\sum_{i=1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} out_1^1 \\ \ldots \\ \frac{1}{n}\sum_{i=1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} out_m^1 \end{bmatrix}$$

$$\frac{\partial E_X}{\partial B^2} = \frac{1}{n}\sum_{i=1}^{n} \left(\frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} \right)$$

Backproparadion: MLP matrix notation

Matrix notation (batch): $X = (x^1 \dots x^n)$ where: $x^i = \left(x_1^i, \dots, x_k^i\right)$

The matrices from the previous slide can be obtained using the formulas:

Exercise: Calculate manually to demonstrate (homework)

$$\frac{\partial E_X}{\partial W^2} = \frac{1}{n} {Out_X^1}^T \times \left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2}\right)$$

$$\frac{\partial E_X}{\partial B^2} = \frac{1}{n} \vec{1} \times \left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2}\right)$$

matrix[n, number of neurons in L2]

matrix[no of neurons in L1, n]

Matrix [1, n]

Backproparadion: MLP matrix notation

Matrix notation (batch): $X = (x^1 \dots x^n)$ where: $x^i = \left(x_1^i, \dots, x_k^i\right)$

The matrices from the previous slide can be obtained using the formulas:

$$\frac{\partial E_X}{\partial W^2} = \frac{1}{n} {Out_X^1}^T \times \left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2}\right)$$

$$\frac{\partial E_X}{\partial B^2} = \frac{1}{n} \vec{1} \times \left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2}\right)$$

$${Out_X^1}^T = \begin{bmatrix} out_1^1(x_1) & \dots & out_1^1(x_n) \\ \dots & \dots & \dots \\ out_m^1(x_1) & \dots & out_m^1(x_n) \end{bmatrix}$$

$$\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2} = \begin{bmatrix} \frac{\partial E(x^1)}{\partial out_1^2(x^1)} * \frac{\partial out_1^2(x^1)}{\partial in_1^2(x^1)} \\ \dots \\ \frac{\partial E(x^n)}{\partial out_1^2(x^n)} * \frac{\partial out_1^2(x^n)}{\partial in_1^2(x^n)} \end{bmatrix}$$

MLP:

Matrix Notation: Batch

LAYER 1

Gradient Optimisation Method: Example

Matrix notation (batch): $X = (x^1 \dots x^n)$ where: $x^i = (x_1^i, \dots, x_k^i)$

$$\frac{\partial E_X}{\partial W^1} = \begin{bmatrix} \frac{1}{n}\sum_{i-1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} w_1^2 \frac{\partial out_1^1(x^i)}{\partial in_1^1(x^i)} x_1^i & \cdots & \frac{1}{n}\sum_{i-1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} w_m^2 \frac{\partial out_m^1(x^i)}{\partial in_m^1(x^i)} x_1^i \\ \vdots & \ddots & \vdots \\ \frac{1}{n}\sum_{i-1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} w_1^2 \frac{\partial out_1^1(x^i)}{\partial in_1^1(x^i)} x_k^i & \cdots & \frac{1}{n}\sum_{i-1}^{n} \frac{\partial E(x^i)}{\partial out_1^2(x^i)} \frac{\partial out_1^2(x^i)}{\partial in_1^2(x^i)} w_m^2 \frac{\partial out_m^1(x^i)}{\partial in_m^1(x^i)} x_k^i \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation (batch): $X = (x^1 \ldots x^n)$ where: $x^i = (x_1^i, \ldots, x_k^i)$

The matrix from the previous slide can be obtained using the formula:

Exercise: Calculate manually to demonstrate (homework)

$$\frac{\partial E_X}{\partial W^1} = \frac{1}{n} X^T \times \left(\left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2} \times W^{2^T} \right) * \frac{\partial Out_X^1}{\partial In_X^1} \right)$$

matrix [n, no of neurons in L1]

matrix [k, n]

matrix [no of neurons in Layer 2,no of neurons in L1]

matrix[n, number of neurons in L2]

Gradient Optimisation Method: Example

Matrix notation (batch): $X = (x^1 \ldots x^n)$ where: $x^i = (x_1^i, \ldots, x_k^i)$

The matrix from the previous slide can be obtained using the formula:

$$\frac{\partial E_X}{\partial W^1} = \frac{1}{n} X^T \times \left(\left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2} \times {W^2}^T \right) * \frac{\partial Out_X^1}{\partial In_X^1} \right)$$

$$X^T = \begin{bmatrix} x_1^1 & \ldots & x_1^n \\ \ldots & \ldots & \ldots \\ x_k^1 & \ldots & x_k^n \end{bmatrix}$$

$$\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2} = \begin{bmatrix} \frac{\partial E(x^1)}{\partial out_1^2(x^1)} * \frac{\partial out_1^2(x^1)}{\partial in_1^2(x^1)} \\ \ldots \\ \frac{\partial E(x^n)}{\partial out_1^2(x^n)} * \frac{\partial out_1^2(x^n)}{\partial in_1^2(x^n)} \end{bmatrix}$$

$$\frac{\partial Out_X^1}{\partial In_X^1} = \begin{bmatrix} \frac{\partial out_1^1(x^1)}{\partial in_1^1(x^1)} & \ldots & \frac{\partial out_m^1(x^1)}{\partial in_m^1(x^1)} \\ \ldots & \ldots & \ldots \\ \frac{\partial out_1^1(x^n)}{\partial in_1^1(x^n)} & \ldots & \frac{\partial out_m^1(x^n)}{\partial in_m^1(x^n)} \end{bmatrix}$$

Gradient Optimisation Method: Example

Matrix notation: (batch): $X = (x^1 \dots x^n)$ where: $x^i = (x_1^i, \dots, x_k^i)$

$$\frac{\partial E_X}{\partial B^1} = \frac{1}{n}\vec{1} \times \left(\left(\frac{\partial E_X}{\partial Out_X^2} * \frac{\partial Out_X^2}{\partial In_X^2} \times {W^2}^T\right) * \frac{\partial Out_X^1}{\partial In_X^1}\right)$$

matrix [n, no of neurons in L1]

matrix [1, n]

matrix [no of neurons in L1,no of neurons in L1]

matrix[n, number of neurons in L2]

Python Programming Example

MLP Backpropagation.ipynb

Next to come:

Different Activation Functions